

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 5 -- Comprehensive (All Topics)

Duration: 90 minutes. Total: 100 points. Show all working.

This exam covers all course topics. Each question integrates multiple areas.

Q1 (30 points). A 4-state Markov chain on {A, B, C, D} models a system's health status. The transition matrix is:

	A	B	C	D
A	[0.6	0.4	0.0	0.0]
B	[0.2	0.3	0.5	0.0]
C	[0.0	0.0	0.7	0.3]
D	[0.0	0.0	0.4	0.6]

- Identify all communicating classes and label each as recurrent or transient. Is the full chain irreducible?
- Find the stationary distribution of the recurrent class. Set up and solve the balance equations restricted to the recurrent states.
- The rate of convergence to stationarity is governed by the second-largest eigenvalue of the transition matrix WITHIN the recurrent class. For the recurrent class with transition matrix P_r , the eigenvalues are 1 and $\lambda(2) = P_r(C,C) + P_r(D,D) - 1 = 0.7 + 0.6 - 1 = 0.3$. After how many steps is the total variation distance to stationarity below 0.01? Use the bound: $d(n) \leq (1/2) * |\lambda(2)|^n$ and solve $(1/2)*(0.3)^n < 0.01$.

Q2 (35 points). A server for a cloud application can be in three states: Idle (I), Processing (P), and Crashed (C). The CTMC rate matrix is:

	I	P	C
I	[-3.0	3.0	0.0]
P	[1.0	-4.0	3.0]
C	[2.0	0.0	-2.0]

- Identify $q(I)$, $q(P)$, $q(C)$ and write the embedded chain P (divide each off-diagonal row entry by the row's total leaving rate).
- Find the stationary distribution π of the CTMC by solving $\pi * Q = 0$ with $\pi(I) + \pi(P) + \pi(C) = 1$.
- The server is 'available' only when in state I or P. Use the Renewal-Reward Theorem to find the long-run fraction of time the server is available. Define reward $R(n)$ = time spent in I or P during cycle n , where a cycle starts at each entry into I.
Alternatively: the fraction of time in I = $\pi(I)$ and in P = $\pi(P)$; compute $\pi(I) + \pi(P)$ directly

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from (b).

Q3 (35 points). A player starts with $k = 3$ chips in a casino game. At each turn the player either gains 1 chip (prob $p = 0.5$) or loses 1 chip (prob $q = 0.5$). The game ends when the player has either 0 chips (ruin) or $N = 7$ chips (win). Let $S(n)$ = chips at turn n .

- Define the three processes: (i) $S(n)$, (ii) $S(n)^2 - n$, and (iii) for a biased walk with $p \neq 1/2$, $(q/p)^{S(n)}$. State (without proof) which of these are martingales and under what conditions on p .
- Apply the OST to martingale $S(n)$ (symmetric walk, $p = 1/2$) to find the win probability $h(3) = P(\text{reach } 7 \text{ before } 0 \mid \text{start } 3)$. Then apply OST to $S(n)^2 - n$ to find $E[T \mid \text{start } 3]$, the expected game length.
- The player considers a modified stopping rule: they stop early if they reach 5 chips even if they haven't reached 7. The new stopping time is $T' = \min\{T, T_5\}$ where T_5 = first time $S(n)=5$. Without calculation, explain why $E[S(T')] = 3$ still holds by OST and what this tells us about $P(\text{reach } 5 \text{ before } 0 \mid \text{start } 3)$.